 Backtesting & Metrics

Advanced Risk-Adjusted Trading Metrics: The Definitive Guide

Sharpe Ratio, Sortino, Calmar, SQN and K-Ratio: what each measures, how to calculate them, and which to prioritize based on your trading profile.

Rubén Villahermosa • January 26, 2026 • 18 min read

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! Which one for your profile?

Your strategy shows 45% annual return. Sounds spectacular. But **how much risk did you take to achieve it?** Did you suffer a 60% drawdown? Would the volatility have knocked you out of the market three times? Without risk-adjusted metrics, you're driving blind.

If you've already mastered fundamental metrics like Net Profit, Win Rate and Profit Factor, it's time to level up to the metrics that truly separate professional traders from amateurs.

> *"Risk-adjusted metrics measure how much return you get for each unit of risk taken. They're the difference between a professional trader and a gambler with temporary luck."*

Calculate all metrics automatically

Algo Strategy Analyzer calculates Sharpe, Sortino, Calmar, SQN and +27 additional metrics with a single click.

Analyze my strategy →

Why basic metrics aren't enough

Basic metrics like Net Profit, Win Rate or Profit Factor only tell half the story. A system with 80% win rate can blow up your account if the losses are catastrophic. A profit factor of 2.0 means nothing if volatility prevents you from holding positions.

The fundamental problem is that **profitability without risk context is incomplete information**.

Example: Two systems, same profitability

- **System A:** +50% annual with 10% max drawdown
- **System B:** +50% annual with 45% max drawdown

Both have the same profitability, but **System A is 4.5 times better** on a risk-adjusted basis.

William Sharpe, 1990 Nobel Prize in Economics, formalized this concept in 1966: **it's not enough to know how much you win, you need to know how much risk you take to achieve it**.

Sharpe Ratio: The most famous metric (and its limitations)

The **Sharpe Ratio** is the industry standard for measuring risk-adjusted performance. Developed by William F. Sharpe, it measures excess return per unit of total volatility.

```
Sharpe Ratio = (Rp - Rf) / σp
```

Rp = Portfolio return (annualized) | **Rf** = Risk-free rate (2-5%) | **σp** = Standard deviation of returns

Practical calculation example

Assume a strategy with:

- Annual return: 25%
- Risk-free rate: 4%
- Annual standard deviation: 15%

```
Sharpe = (25% - 4%) / 15% = 21% / 15% = 1.40
```

A Sharpe of 1.40 indicates that for each percentage point of volatility, you get 1.40 points of excess return.

Value interpretation

Sharpe Ratio	Interpretation
< 0	Return below risk-free rate
0 - 0.5	Poor
0.5 - 1.0	Acceptable
1.0 - 2.0	Good (professional standard)
2.0 - 3.0	Very good
> 3.0	Excellent (suspicious if sustained)

Critical limitations of the Sharpe Ratio

The fundamental flaw of Sharpe

The Sharpe Ratio **treats upside volatility the same as downside volatility**. If your strategy has occasional large gains (typical of trend following systems), Sharpe unfairly penalizes it.

Other important limitations:

1. Assumes normal distribution of returns

Markets have "fat tails" (extreme events more frequent than normal curve predicts). Sharpe underestimates real risk.

2. Ignores the order of returns

A Sharpe of 1.5 can come from consistent returns or a roller coaster that ends well.

3. Sensitive to calculation period

The same system can show Sharpe 2.0 in one period and 0.8 in another.

4. Doesn't capture ruin risk

A system can have good Sharpe but expose you to catastrophic losses in extreme scenarios.

RATIO

Sortino Ratio: Measuring only the risk that matters

So

The **Sortino Ratio** was developed by Frank A. Sortino in the 1980s to correct Sharpe's main flaw: penalizing upside volatility.

The idea is simple but powerful: **investors don't worry about winning "too much", they only worry about losing**.

```
Sortino Ratio = (Rp - Rf) / σd
```

σd = Downside Deviation (deviation of negative returns only)

Downside Deviation: The key to Sortino

Downside Deviation considers only returns below a threshold (usually 0% or the risk-free rate):

```python
import numpy as np

def downside_deviation(returns, threshold=0):
    negative_returns = returns[returns < threshold]
    if len(negative_returns) == 0:
        return 0
    return np.sqrt(np.mean(negative_returns**2))
```

Comparative example: Sharpe vs Sortino

Consider two strategies:

Metric	Strategy A (Mean Reversion)	Strategy B (Trend Following)
Annual return	~12%	~25%
Sharpe Ratio	**5.7**	4.1
Sortino Ratio	8.9	**17.5**

Sharpe favors Strategy A because it has lower total volatility. But **Sortino recognizes that Strategy B is superior**: it has higher return and its volatility comes mainly from large gains, not losses.

When to prefer Sortino over Sharpe

- Your strategy is trend following type
- You have asymmetric return distribution (few large winning trades)
- You want a more realistic measure of perceived risk

The consensus among institutional traders is that Sortino is more useful than Sharpe for most real strategies.

RATIO C

Calmar Ratio: The worst-case scenario as reference

The **Calmar Ratio** was developed by Terry W. Young in 1991 and takes a different approach: instead of measuring volatility, it measures performance relative to the strategy's worst moment.

```
Calmar Ratio = CAGR / |Max Drawdown|
```

CAGR = Compound annual growth rate | **Max Drawdown** = Maximum peak-to-trough decline

Practical example

A strategy with 20% CAGR and -25% Max Drawdown:

```
Calmar = 20% / 25% = 0.80
```

A Calmar of 0.80 means that for each 1% of maximum drawdown suffered, you got 0.80% of annualized return.

Value interpretation

Calmar Ratio	Interpretation
< 0.5	Poor return/drawdown ratio
0.5 - 1.0	Acceptable
1.0 - 2.0	Good
2.0 - 3.0	Very good
> 3.0	Excellent

Reference data

- **Typical 60/40 portfolio:** Calmar 0.8-1.2
- **Bitcoin 2020-2025:** Sharpe ~0.95, Sortino 1.93, Calmar 0.84
- **Professional CTA strategies:** Typical Calmar 1.0-2.0

Want to see these metrics in action?

Import your TradingView, MetaTrader or TradeStation history and get a complete diagnosis with all advanced metrics.

Try for free →

SQN (System Quality Number): Van Tharp's vision

The **SQN (System Quality Number)** was created by Van K. Tharp in 2008 in his book "The Definitive Guide to Position Sizing". It's a unique metric because it combines expectancy, consistency and opportunities into a single number.

$$SQN = \sqrt{N} \times Mean(R) / StdDev(R)$$

N = Number of trades | **R** = R-Multiples (gain as multiple of initial risk)

Understanding R-Multiples

An R-Multiple expresses each trade in terms of your initial risk:

- If you risk $100 and win $200, your R = **+2R**
- If you risk $100 and lose $100, your R = **-1R**
- If you risk $100 and win $50, your R = **+0.5R**

Van Tharp's interpretation scale

SQN	Rating
1.6 - 1.9	Below average, but tradeable
2.0 - 2.4	Average
2.5 - 2.9	Good
3.0 - 5.0	Excellent
5.1 - 6.9	Superb
7.0+	"Maybe you've got the Holy Grail"

What makes SQN special

SQN captures three critical dimensions:

💰

Expectancy

How much do you win per trade on average?

📊

Consistency

How predictable are your results?

🔄

Opportunities

How many times can you apply your edge?

⚠️

Important limitation

SQN requires **minimum 30 trades** to be statistically significant. Also, it favors mean reversion strategies over trend following.

K-Ratio: The consistency of your equity curve

The **K-Ratio** was developed by Lars N. Kestner in 1996 and published in Technical Analysis of Stocks & Commodities. It measures something other metrics ignore: **the linearity of your equity curve**.

```
K-Ratio = Slope / Standard Error of Slope
```

The slope is calculated via linear regression on the cumulative equity curve.

Why the K-Ratio is unique

Other metrics (Sharpe, Sortino, Calmar) are **insensitive to the order of returns**. The K-Ratio detects this:

Example: Same result, different path

Sequence A: +5%, +5%, +5%, -15%, +5%, +5%

Sequence B: +5%, -15%, +5%, +5%, +5%, +5%

Both have same total return, same standard deviation, and same max drawdown. Therefore, **same Sharpe, Sortino and Calmar**. But Sequence B has **worse K-Ratio** because the large loss at the beginning creates a less linear curve.

Python calculation

```
import numpy as np
from scipy import stats

def k_ratio(equity_curve):
```

```
    """Calculates the K-Ratio of an equity curve"""
    n = len(equity_curve)
    x = np.arange(n)

    # Linear regression
    slope, intercept, r_value, p_value, std_err = stats.linregress(x, equity_curve

    # K-Ratio = slope / standard error
    if std_err == 0:
        return float('inf')

    return slope / std_err
```

A K-Ratio above 2.0 is considered good because it indicates an ascending and relatively smooth equity curve. The K-Ratio is especially valuable when creating an algorithmic strategy, as it helps detect overfitting before going live.

MORE +

Other advanced metrics you should know

Beyond the five core metrics, there are other risk-adjusted metrics that hedge funds and institutional traders use frequently. Knowing them will give you a more complete perspective when evaluating the anatomy of a trading strategy.

Omega Ratio

The **Omega Ratio** is one of the most comprehensive metrics because it considers the entire return distribution, not just the mean and variance. Unlike Sharpe, it does not assume normal distribution and correctly captures the "fat tails" (leptokurtic distributions) typical of financial markets.

```
Omega Ratio = Sum(returns > threshold) / Sum(returns <
threshold)
```

Ratio of the weighted sum of gains over the weighted sum of losses relative to a threshold (usually 0%).

An Omega Ratio above 1.0 indicates that weighted gains exceed weighted losses. The higher, the better. Unlike Sharpe, Omega captures information from the entire distribution, including asymmetries and fat tails.

Ulcer Index

The **Ulcer Index** was created to literally measure the "pain" an investment causes. While standard deviation measures total volatility, the Ulcer Index focuses exclusively on drawdowns and their duration.

```
Ulcer Index = sqrt(Mean of Squared Drawdowns)
```

Measures the accumulated pain from drawdowns. A lower value means less drawdown suffering.

The Ulcer Index is especially relevant for conservative investors and wealth managers, because it reflects the real investor experience during loss periods. A low Ulcer Index (for example, below 5) indicates that the strategy generates little psychological stress.

MAR Ratio

The **MAR Ratio** (Managed Account Reports Ratio) is functionally similar to the Calmar Ratio but is typically calculated over a longer period (the entire life of the strategy, instead of the last 3 years used by Calmar). Its formula is identical:

```
MAR Ratio = CAGR / |Max Drawdown|
```

Same as Calmar but calculated over the full life of the account or strategy, not just the last 36 months.

The key difference is temporal: Calmar is typically calculated over 36 months (rolling), while the MAR Ratio spans the entire history. This makes it more stable but less reactive to recent changes.

Information Ratio and Treynor Ratio

Two additional metrics of institutional origin complete the professional manager's arsenal:

Information Ratio

Measures a portfolio's excess return relative to its benchmark, divided by the tracking error (deviation of the return difference). It is the preferred metric for evaluating active managers against their reference index.

Treynor Ratio

Similar to Sharpe but uses beta (systematic risk) instead of standard deviation (total risk). Its formula is: (Rp - Rf) / Beta. It is useful when the portfolio is part of a diversified portfolio, as it only measures non-diversifiable risk.

Summary table: All advanced metrics

Metric	Formula	Key advantage
Omega Ratio	`Sum(R > threshold) / Sum(R < threshold)`	No normal distribution assumption, captures fat tails
Ulcer Index	`sqrt(Mean(DD squared))`	Measures real pain of drawdowns, not just magnitude
MAR Ratio	`CAGR / |Max DD|`	Long-term view of return vs drawdown
Information Ratio	`(Rp – Rb) / Tracking Error`	Evaluates active management vs benchmark
Treynor Ratio	`(Rp – Rf) / Beta`	Systematic risk only (non-diversifiable)

Comparison table: All metrics at a glance

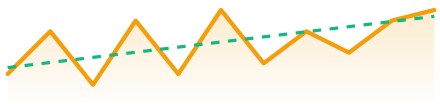
Metric	What it measures	Optimal value	Best for
Sharpe Ratio	Return / Total volatility	> 1.0 pro, > 2.0 excellent	General comparison
Sortino Ratio	Return / Downside volatility	> Sharpe always	Trend following
Calmar Ratio	Return / Max Drawdown	> 1.0 good, > 2.0 very good	Drawdown-averse
SQN	Systemic quality	> 2.5 good, > 3.0 excellent	Holistic evaluation
K-Ratio	Equity curve linearity	> 2.0 good	Temporal consistency

Warning Signs: When a Metric Lies

Each metric has blind spots. **A single metric can deceive you, but the combination reveals the truth.** Here are 4 real scenarios where one metric would lead you to an incorrect decision:

High Sharpe + Low K-Ratio

The erratic curve that averages well



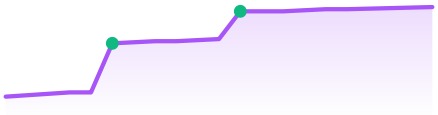
1.45
Sharpe ✓

0.6
K-Ratio ✕

What it hides: Sharpe only sees the final average. K-Ratio detects that the equity jumped up and down chaotically. **This system may have been lucky.**

Low Sharpe + High Sortino

The profitable but volatile system



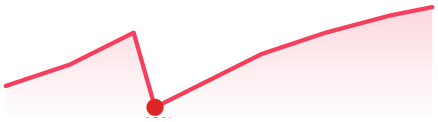
0.85
Sharpe ✕

2.80
Sortino ✓

What it hides: Sharpe penalizes the large winning trades. Sortino reveals that volatility is upward, not downward. **Typical of trend following systems.**

All OK + Low Calmar

The hidden devastating drawdown



1.30
Sharpe ✓

0.45
Calmar ✕

What it hides: Sharpe sees "average volatility". Calmar reveals that a single 48% drawdown could have ruined you. **Could you have held on?**

Excellent SQN + Few Trades

The statistically insignificant result

4.2
SQN ✓

18
Trades ✕

What it hides: SQN of 4.2 sounds excellent, but with only 18 trades it has no statistical validity. **Van Tharp requires minimum 30 trades.**

🏆

The Golden Rule

Never trust a single metric. **A robust strategy passes multiple filters:**

- ✅ Sharpe > 1.0 (minimum professional quality)
- ✅ Sortino > Sharpe (positive volatility is not bad)
- ✅ Calmar > 0.8 (acceptable return/drawdown)
- ✅ K-Ratio > 1.5 (consistent equity curve)
- ✅ SQN > 2.0 with 30+ trades (statistical significance)

If it fails on two or more, investigate before risking real capital.

CHOOSE !

Which to prioritize based on your trading profile?

There's no universally better metric. The choice depends on your style, risk tolerance and objectives.

Conservative trader (capital preservation)

Primary metric: Calmar Ratio

If your priority is not losing, Calmar tells you exactly how much you can expect to earn per point of drawdown.

Trend follower

Primary metric: Sortino Ratio

Trend following strategies have few but very large winning trades. Sortino recognizes that this "positive" volatility is desirable.

High-frequency trader (scalper)

Primary metric: SQN

With many small trades, you need a metric that values consistency and number of opportunities.

Fund manager / Institutional

Primary metric: Sharpe Ratio

Sharpe remains the industry standard for reporting to investors and allows comparison with benchmarks.

System developer

Primary metric: K-Ratio + SQN

When evaluating and optimizing systems, you want to avoid overfitting. K-Ratio detects suspiciously "perfect" curves.

Summary table: Metric by profile

For a quick reference, this table condenses which metric to prioritize based on your trading style and the main reason behind that choice:

Profile	Primary metric	Reason
Conservative (capital preservation)	Calmar Ratio	Measures return per point of drawdown suffered
Trend follower	Sortino Ratio	Doesn't penalize large gains as volatility
Scalper / High frequency	SQN	Values consistency and number of opportunities
Fund manager / Institutional	Sharpe Ratio	Industry standard for reporting to investors
System developer	K-Ratio + SQN	Detects suspicious curves and overfitting

REAL !

Reality vs theory: What they don't tell you

Numbers can lie

A Sharpe Ratio of 3.0 on a 2-year backtest probably won't survive live. Metrics are sensitive to:

- **Selected period:** Cherry-picking favorable dates
- **Excluded costs:** Slippage, commissions, spread
- **Overfitting:** Parameters optimized for the past (see backtest problems)

📉

Degradation is inevitable

According to systematic strategy studies, **live Sharpe Ratio is typically 30-50% lower than backtest**. Apply this "haircut" to your expectations.

Combined metrics > Single metric

Professional traders never look at just one metric. A solid strategy should pass multiple filters:

Minimum metrics checklist

- ✅ Sharpe > 1.0
- ✅ Sortino > 1.5
- ✅ Calmar > 0.8
- ✅ SQN > 2.0 (if enough trades)
- ✅ K-Ratio > 1.5

If it fails on two or more, there are warning signs.

The most important metric isn't a number: Can you execute this system consistently? A system with Sharpe 1.2 that you can follow with discipline is infinitely better than one with Sharpe 2.5 that

will knock you out of the market at the first drawdown. To ensure your strategy withstands full scrutiny, check our guide to validate your strategy step by step.

Conclusion

Risk-adjusted metrics transform raw performance data into actionable information. But no metric is perfect or universal.

The 3 key takeaways

1. **Sharpe Ratio is the standard but has limitations.** It penalizes large gains and assumes normal distribution. Use Sortino as a complement.
2. **Choose metrics based on your profile.** Conservative → Calmar. Trend followers → Sortino. High frequency → SQN. Developers → K-Ratio.
3. **Never trust a single metric.** Robust strategies pass multiple filters. If one metric is excellent but another is terrible, investigate.

Metrics are tools, not answers. They help you ask better questions about your trading.

Calculate all metrics automatically

Import your TradingView, MetaTrader or TradeStation history and get Sharpe, Sortino, Calmar, SQN and +27 additional metrics.

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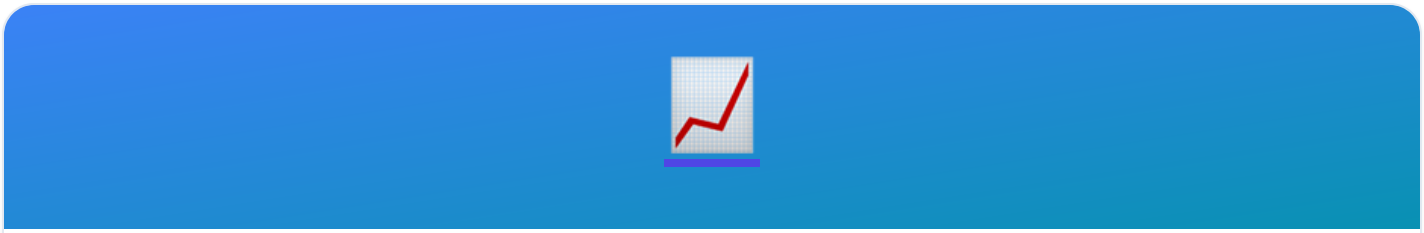

Fundamental Metrics

Net Profit, Win Rate, Profit Factor and more

Backtesting Problems

7 biases that make your backtest lie

Validate Your Strategy

Complete step-by-step validation guide

FAQ ?

Frequently Asked Questions

- What is a good Sharpe Ratio value?
- Why is Sortino better than Sharpe for most traders?
- How many trades do I need to calculate SQN?
- Can a strategy have good Sharpe but bad K-Ratio?
- What metric do hedge funds use?
- Is it possible to have a negative SQN?
- How does the risk-free rate affect Sharpe and Sortino?
- How often should I recalculate these metrics?

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